



UNIVERSITY OF GHANA

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**MPhil./MSc. COMPUTER SCIENCE, EXAMINATION FOR FIRST SEMESTER:
2022/2023**

CSCD 603: ADVANCED ALGORITHM AND DATA STRUCTURES (3 CREDITS)

INSTRUCTION:

Please ensure that all your answers are brief and properly demonstrate your understanding of the concepts. Marks will be awarded for clarity of expression. Answer all questions in the answer booklet provided.

Answer All Question

TIME ALLOWED:

THREE (3) HOURS

ANSWER ALL QUESTION

1. a) What is the matrix-chain multiplication problem and from your definition what will be the minimum cost and the order of parenthesis of $A_1A_2A_3$ if $A_1=[30,35]$, $A_2=[35,15]$ and $A_3=[15,5]$ [8marks]
- b) Let $R(i,j)$ be the number of times that table entry $m[i,j]$ is referenced while computing other table entries in a call of MATRIX-CHAIN-ORDER (*see supplementary sheet*). Show that the total number of references for the entire table [6marks]

$$\sum_{i=1}^n \sum_{j=i}^n R(i,j) = \frac{n^3 - n}{3}$$

- c) In what two ways do optimal substructure in dynamic programming varies across problem domains? [2marks]

d) What is the potential value of a Fibonacci Heap with 40 nodes where 10 out of the 40 are in the root list while 31 out of the 40 have their marked attribute set to FALSE [2marks]

e) Summarize in three sequence of steps the design of a greedy algorithm [8marks]

f) Explain briefly the greedy-choice property as a key ingredient of greedy algorithm [4marks]

Total: [30 marks]

2. a) In what ways are B-Tree different or similar to Red-Black trees [4marks]

b) Given a B-Tree whose root node contains 1000 keys, how many nodes and keys are expected on the second level. Here the second level is defined as a tree with height two.

[4marks]

c) State and explain with examples the three basic operations of B-Tree [8marks]

d) Explain in details the structure of a Fibonacci Heap [4marks]

Total: [20 marks]

3. a) The following pseudocode deletes a node from an n-node Fibonacci heap. From the structure given delete the now 21. The auxiliary functions in this pseudocode is found on the supplementary sheet. [20marks]

FIB-HEAP-DELETE(H, x)

1 FIB-HEAP-DECREASE-KEY($H, x, -\infty$)

2 FIB-HEAP-EXTRACT-MIN(H)

Total: [20 marks]

Supplementary Note

MATRIX-CHAIN-ORDER()

FIB-HEAP-DECREASE-KEY()

FIB-HEAP-EXTRACT-MIN()